

Ganglia

Ganglia is a distributed monitoring system designed for monitoring large clusters and HPC environments.

Step-1: Install the package

- Install in both nodes

```
# sudo yum install epel-release -y
```

```
# sudo yum install ganglia-gmond -y
```

- Install only in master (central server)

```
# sudo yum install ganglia-gmetad ganglia-web rrdtool httpd -y
```

Step-2: configure the gmond (unicast mode)

On both node

```
# sudo vi /etc/ganglia/gmond.conf
```

- Comment the multicast

```
# udp_send_channel {  
#   mcast_join = 239.2.11.71  
#   port = 8649  
#   ttl = 1  
# }
```

- Add or change like this

```
udp_send_channel {
```

```
    host = 10.10.40.88
```

```
port = 8649
```

```
}
```

```
udp_send_channel {  
    host = 10.10.40.88  
    port = 8649  
}
```

```
udp_recv_channel {
```

```
port = 8649
```

```
}
```

```
udp_recv_channel {  
    port = 8649  
}
```

Only on slave nodes

```
" cluster {
```

```
name = "cluster"
```

```
}"
```

```
cluster {  
    name = "cluster"  
}
```

On both nodes

```
# sudo systemctl enable gmond
```

```
# sudo systemctl restart gmond
```

Step-3: configure metad

Only on master node

```
# sudo vi /etc/ganglia/gmetad.conf
```

- Find and change this

```
" data_source "cluster" 10.10.40.88 "
```

```
data_source "cluster" 10.10.40.88
```

```
# sudo systemctl enable gmetad
```

```
# sudo systemctl restart gmetad
```

Step-4: configure web UI (master node)

```
# ls /var/www/html/
```

- Check if there is any ganglia dir. Is present. If not follow these steps

```
# sudo ln -s /usr/share/ganglia /var/www/html/ganglia
```

```
# sudo vi /etc/httpd/conf.d/ganglia.conf
```

- Modify the apache permission
- It will show like require local and change like this

```
" <Directory "/usr/share/ganglia">
```

```
    Require all granted
```

```
</Directory> "
```

```
# sudo systemctl enable httpd
```

```
# sudo systemctl start httpd
```

Step-5: Firewall and selinux (only on master node)

```
# sudo firewall-cmd --permanent --add-service=http

# sudo firewall-cmd --permanent --add-port=8649/udp

# sudo firewall-cmd --reload

# sudo getenforce

# sudo setenforce 0
```

Step-6: Check in browser: <http://10.10.40.88/ganglia>

Troubleshooting

- Check apache on master node

```
# sudo systemctl status httpd

# sudo systemctl enable httpd

# sudo systemctl start httpd

# curl http://localhost
```

- Check firewall

```
# sudo firewall-cmd --list-all

# sudo firewall-cmd --permanent --add-service=http

# sudo firewall-cmd --permanent --add-port=8649/udp

# sudo firewall-cmd --reload
```

- Check selinux

```
# getenforce. Check it is Permissive
```

```
# setenforce 0
```

- Verify Ganglia Web Directory

```
# ls /var/www/html/
```

```
# sudo ln -s /usr/share/ganglia /var/www/html/ganglia
```

```
# sudo systemctl restart httpd
```

- Test From Master node Itself

```
# curl http://localhost/ganglia
```

- Check Ganglia Web Config

```
# sudo vi /etc/httpd/conf.d/ganglia.conf
```

- Modify the apache permission
- It will show like require local and change like this

```
" <Directory "/usr/share/ganglia">
```

```
    Require all granted
```

```
</Directory> "
```

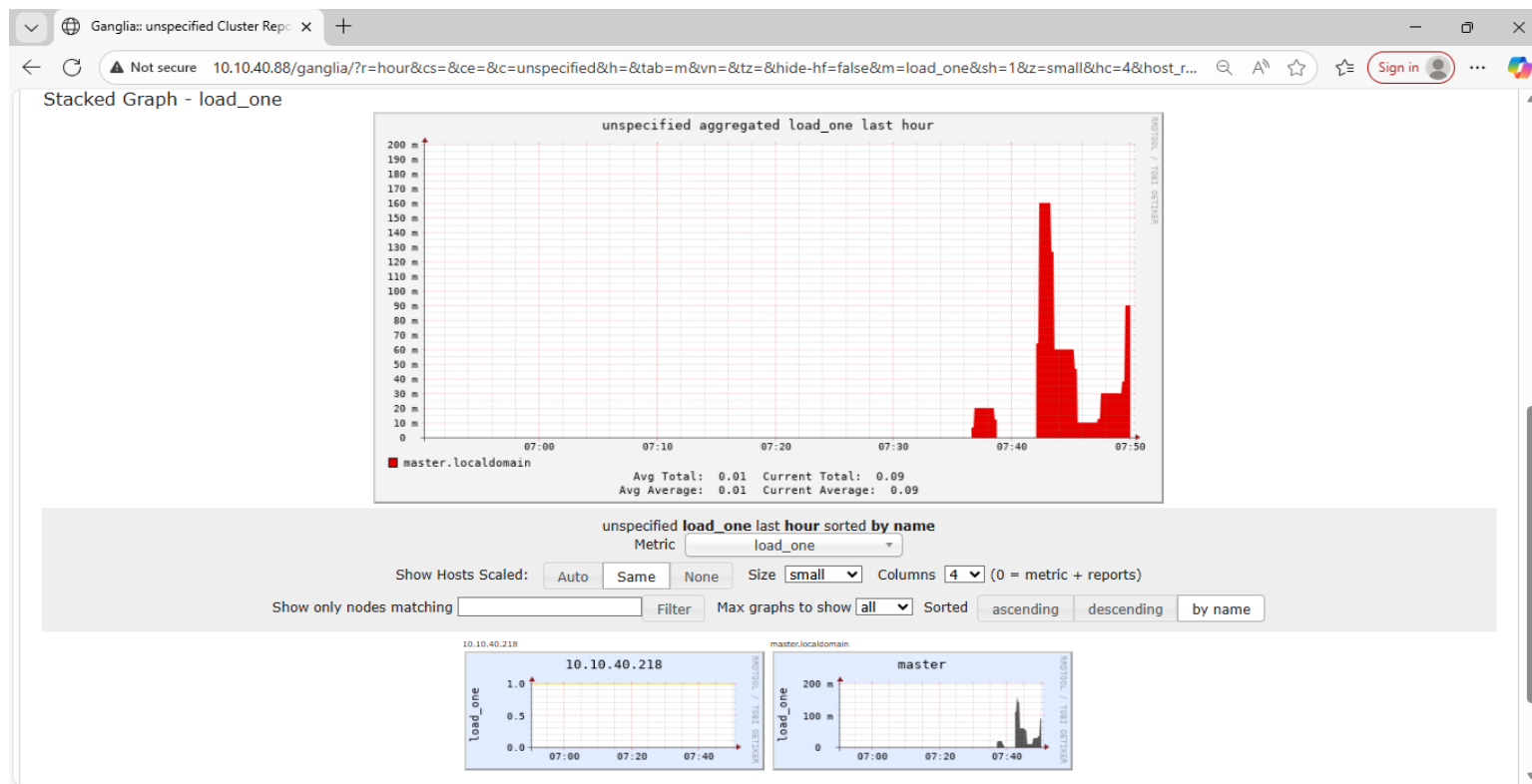
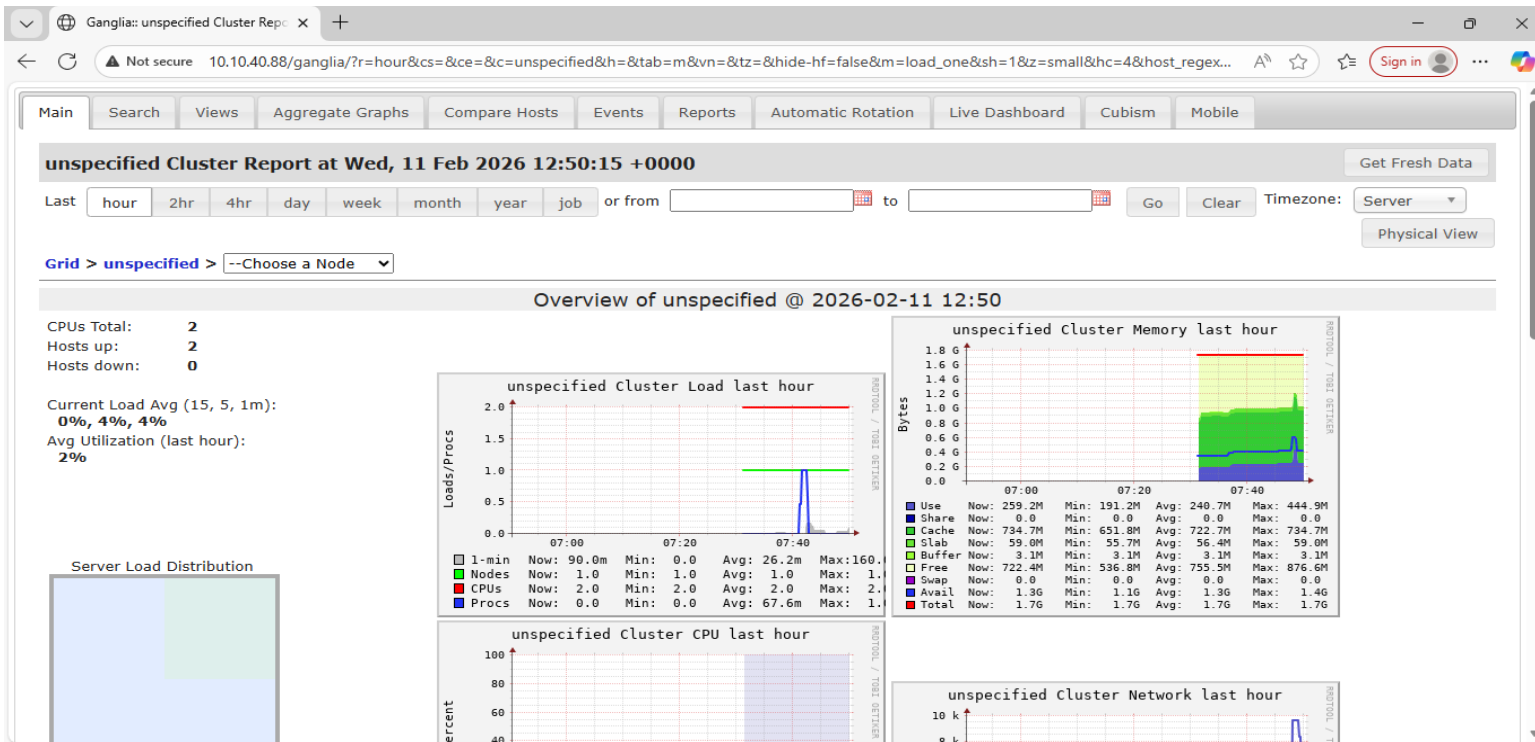
```
# sudo systemctl restart httpd
```

- If it shows Forbidden

Problem is with Apache config still has Require local or SELinux blocking

- Check in browser: <http://10.10.40.88/ganglia>

Output:



Monitor Switch & Router in Ganglia

Ganglia cannot directly monitor switches and routers by default because it works using gmond agent installed on servers.

Monitor switch and router in Ganglia Flow

Switch/Router → SNMP → SNMP Collector → Ganglia

Services can be used in ganglia

Ganglia is mainly designed for server & cluster monitoring, not network devices.

Let me clearly show you what all you can monitor and use with Ganglia. Like:

1. System / Server Monitoring (Main Purpose)
 - CPU, Memory, Disk, Network, System
2. Cluster Monitoring (HPC / Lab)
 - Ganglia is very powerful for: Monitoring 10, 50, 100+ nodes, Comparing load across servers, Finding overloaded machines, Visualizing cluster performance
3. Custom Metrics (Very Powerful Feature)
 - You can push **custom metrics** using gmetric.
4. Application Monitoring
 - You can monitor: Apache, Nginx, MySQL, Docker, Kubernetes nodes (basic level)
5. Hardware Monitoring
 - If supported: CPU temperature, Fan speed, Power usage, RAID status
6. Alerting (Limited)
 - Ganglia does NOT have strong built-in alerting.
7. Capacity Planning

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Procedure to be follow as per ChatGPT*

Method to Monitor Switch & Router in Ganglia  Step 1 — Enable SNMP on Router/Switch

Example (Cisco style):

```
snmp-server community public RO
```

For other vendors (HP, MikroTik, etc.), enable SNMP in settings.

 Step 2 — Install SNMP Tools on Rocky Linux (Master Node)

On your Ganglia master:

```
sudo dnf install net-snmp net-snmp-utils -y
```

Test SNMP:

```
snmpwalk -v2c -c public 10.10.40.50
```

(Replace with router IP)

If it returns data → SNMP working.

 Step 3 — Install gmetric Tool

Ganglia has tool called gmetric.

It allows you to manually push custom metrics into Ganglia.

Example:

```
gmetric -n switch_cpu -v 25 -t int32 -u "%"
```

This will show as metric in Ganglia UI.

 Step 4 — Create Script to Pull SNMP Data

Example script (CPU usage from router):


```
#!/bin/bash
```

```
CPU=$(snmpwalk -v2c -c public 10.10.40.50 1.3.6.1.x.x.x | awk '{print $NF}')
```

```
gmetric -n router_cpu -v $CPU -t int32 -u "%"
```

Make executable:

```
chmod +x monitor_router.sh
```

Add cron job:

```
crontab -e
```

Run every 1 minute:

```
* * * * * /root/monitor_router.sh
```

Now router metrics will appear in Ganglia.